

MLS Winner Prediction and Feature Importance Report

This report summarizes the process of predicting the Major League Soccer (MLS) season winner using a machine learning model built in this notebook.

1. Data Preprocessing

The dataset "mls_matches.csv" was loaded. Teams were divided into Eastern and Western conferences. Target variables were created for Home Win (H), Draw (D), and Away Win (A). We stacked the data so each row represents a team's perspective.

2. Feature Engineering

Several rolling statistics were created per team, shifted by one match to prevent data leakage. These included points per game (last 5 & 10 matches), win rate, goals scored/conceded, home/away points per game, season points, and league position. An Elo rating system was also implemented to measure team strength.

3. Modeling & Evaluation

Four models were tested: Logistic Regression, Random Forest, Gradient Boosting, and CatBoost. Data was split chronologically (train: < 2025, test: 2025). Logistic Regression was chosen as the best model based on the lowest log loss.

4. Feature Importance

Permutation importance revealed that the difference in "season points" and "win rate over the last 10 games" were the most critical features for predicting match outcomes. League position was also a strong predictor.

5. Prediction & Simulation

The model was retrained on all available data (including early 2026 matches). We simulated the remaining 2026 season 10,000 times to calculate the probability of each team winning the Supporters' Shield (best regular season record) and the MLS Cup (playoff winner).

Predicted Winners (2026)

Supporters' Shield: Inter Miami CF

Inter Miami CF is the overwhelming favorite to win the Supporters' Shield with an estimated 75.4% chance.

MLS Cup: Inter Miami CF

Inter Miami CF is also the favorite to win the MLS Cup, but with a lower probability (30.2%) due to the unpredictable nature of the knockout playoffs. Vancouver Whitecaps (16.3%) and Los Angeles FC (16.0%) are the closest challengers.